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Organizing Committee
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Dear Editor and Organizing Committee,

We are pleased to submit our manuscript titled **“Forecast Driven Communication Scheduling for Koopman Based Multirobot Formation Control”** for consideration at CACS2026, under Special Session SS-17 (Data-Driven Control and Learning for Autonomous Mobile Systems).

The paper proposes a decentralized formation controller in which each robot identifies its own dynamics online through streaming Extended Dynamic Mode Decomposition, plans with a successively linearized batch linear-quadratic model predictive controller, and decides when to broadcast its state using a forecast rather than a reactive criterion. Rather than waiting for a realized state mismatch, each robot rolls its own just-updated Koopman model forward and transmits only when the predicted divergence from its neighbors’ beliefs is about to exceed a safety margin.

On a four-robot formation-tracking task under abrupt, unknown actuation degradation, the proposed rule cuts formation-tracking RMSE by 60.8% relative to reactive triggering at a matched or lower communication budget, and the advantage persists under packet loss, three disturbance profiles, and moderate network delay. A 2×2 ablation study isolates the source of the improvement, showing it traces mainly to the predictive trigger rule itself rather than to online identification alone. We support the empirical results with a proposition bounding the achievable inter-broadcast error, a feature-by-feature comparison against twelve related methods, and a weighted communication-accuracy objective.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission to CACS2026. The authors declare no conflicts of interest. Simulation code and results supporting the paper are publicly available at <https://github.com/hasib123333/Forecast-Driven-Communication-Scheduling>.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

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